

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: _____

May 6, 2017 Saturday

Department/School: _____

SID: _____

University Physics I

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1.(20pts.) A body of mass m is projected upwards with initial speed u and moves under uniform gravity and in a medium that exerts a resistance force that is proportional to the square of its speed. When the sphere falls vertically in the medium, its terminal velocity is v_0 .

(a) Write the equation of the upward motion of the body.

(b) Find the maximum height the body attains and the time t_m taken to reach the maximum height.

(c) Write the equation of the downward motion of the body.

(d) Show that when the body returns to the starting point, its speed is $uv_0/\sqrt{u^2 + v_0^2}$.

(Hint: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$.)

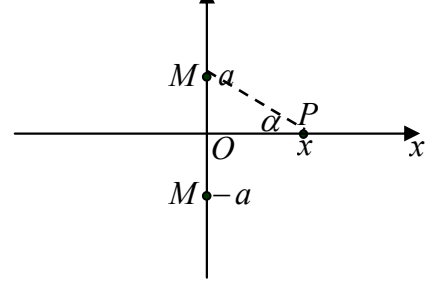
2. (20pts.) A particle P of mass m is confined to move along the x -axis and is under the combined gravitational attraction of two particles, each of mass M , located at the points $(0, a, 0)$ and $(0, -a, 0)$ respectively.

(a) Show that the force field acting on P is given by

$$F(x) = -\frac{2GMmx}{(a^2 + x^2)^{\frac{3}{2}}}.$$

(b) Find the corresponding potential energy function $V(x)$ of the force field of (a).

(c) Find the finite equilibrium point(s) of the particle P , and identify it/them as stable or unstable.



3. (20pts.) A uniform stick of length $2a$ has its lower end in contact with the smooth horizontal surface of a table. Initially the stick stands vertically and is at rest. With a tiny kick the stick topples over.

(a) What is the angular velocity of the stick when it hits the surface of the table?

(b) What is the center-of-mass velocity of the stick when it hits the surface of the table?

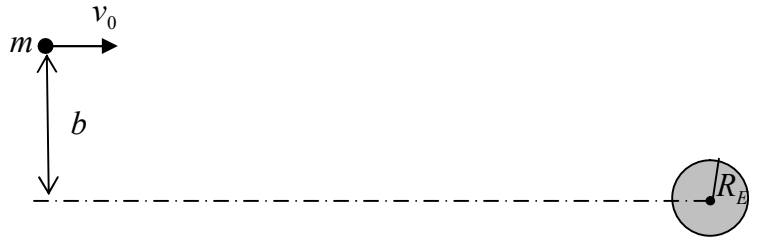
4. (20pts.) An asteroid of mass m approaches the Earth with impact parameter b and velocity v_0 that is relative to the Earth. The influences of the Sun and Moon on the asteroid can be ignored.

(a) What is the initial angular momentum of asteroid with respect to the center of the Earth?

(b) What is the closest approach r_A of the asteroid to the center of the Earth? Express the answer in

terms of b , v_0 , R_E and g where R_E is the radius of the Earth and g is the gravity on the surface of the Earth.

(c) If $b = 20R_E$, what is the condition for v_0 so that the asteroid will hit the Earth?



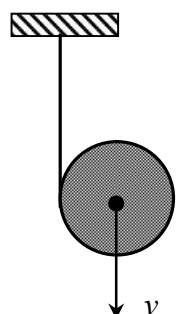
5. (20 pts.) A uniform cylinder (a yo-yo) of radius R has a light inextensible string wrapped around it so that it does not slip. The free end of the string is secured to a fixed point and the yo-yo descends in a vertical straight line.

(a) Write the equation of translational motion of the center of mass and the equation of the rotational motion of the cylinder.

(b) Find the center-of-mass acceleration.

(c) If the cylinder is released from rest and it descends vertically down a height H , what are the center-of-mass velocity and the angular velocity of the cylinder?

(d) Deduce the results of (c) by adopting the conservation of mechanical energy approach.



6. (20pts.) As shown in the figure, a light elastic spring of relaxed length l and spring constant k lies on a smooth horizontal table with one end fixed and the other attached to the left end of a heavy uniform flexible rope of mass M and length $4l$. The other end of the rope hangs down over the edge of the table. When the spring has its relaxed length, the free end of the rope hangs a distance l vertically below the level of the table top. The system is released from rest in this position. The spring constant is $k = Mg/2l$, where g is the gravity.

(a) What is the equilibrium position of the free end of the rope?

(b) Show that the free end of the rope executes simple harmonic motion.

(c) Find the period of the simple harmonic motion of the free end of the rope.

